Wave Propagation and Transmission

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Outline

Chapter Overview

This chapter examines the fundamental principles of electromagnetic wave propagation and transmission. We begin with plane waves—the basic solutions to Maxwell equations—covering their angular-spectrum representation and polarization states. Wave behavior in unbounded media is then analyzed for both lossless and lossy cases. This leads to the study of oblique incidence on dielectric interfaces, including reflection and transmission, the critical angle, and Brewster's angle. The chapter concludes with guided-wave propagation in waveguides, outlining solution methods, mode characteristics, and the use of Green's identities and modal orthogonality.

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The Plane Wave Solution

Plane waves are the most fundamental solutions to the homogeneous Helmholtz equation (1.4.16) in Cartesian coordinates. The analysis begins with the Laplacian operator:

$$\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2. \tag{2.1.1}$$

Using the method of separation of variables, we first seek an elementary solution of the form

$$\psi(x, y, z) = X(x)Y(y)Z(z). \tag{2.1.2}$$

In general, the complete solution can be expressed as a weighted summation or integral of these elementary solutions, depending on whether the spectrum of allowable separation constants is discrete or continuous. Substituting this into the Helmholtz equation yields:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0. \tag{2.1.3}$$

For this equation to be valid for all x, y, z, each term must be a constant. This requirement separates the partial differential equation into three ordinary differential equations:

$$\frac{d^2\Psi}{d\xi^2} + k_\xi^2\Psi = 0, \tag{2.1.4}$$

The Plane Wave Solution (cont.)

where $\Psi \in \{X, Y, Z\}$ and $\xi \in \{x, y, z\}$. The separation constants must satisfy the relation:

$$k_x^2 + k_y^2 + k_z^2 = k^2. \tag{2.1.5}$$

These constants form the components of the wavevector, defined as:

$$\vec{k} = k_x\hat{x} + k_y\hat{y} + k_z\hat{z} = k\hat{k}. \tag{2.1.6}$$

For $k \neq 0$, the notation $\hat{k} = \vec{k}/k$ uses the same chosen scalar branch that satisfies $k^2 = \vec{k} \cdot \vec{k} = k_x^2 + k_y^2 + k_z^2$. It therefore obeys the algebraic, non-conjugated normalization $\hat{k} \cdot \hat{k} = 1$. When $\vec{k}$ is real, $\hat{k}$ is an ordinary geometric unit vector; when $\vec{k}$ is complex, it is a normalized complex direction and need not represent one real geometric direction. The elementary solution is a complex exponential, representing a plane wave:

$$\psi = e^{\pm i(k_x x + k_y y + k_z z)} = e^{\pm i\vec{k}\cdot\vec{r}}. \tag{2.1.7}$$

For a real wavevector and the $e^{i\omega t}$ convention, the time-domain phases are $\omega t \mp \vec{k} \cdot \vec{r}$. Thus $e^{-i\vec{k}\cdot\vec{r}}$ represents propagation in the $+\vec{k}$ direction, while $e^{i\vec{k}\cdot\vec{r}}$ represents propagation in the $-\vec{k}$ direction. (For a complex wavevector, phase and attenuation directions must instead be read from its real and imaginary parts, as done in

Section 2.2.1.) For the remainder of this text, we adopt the former spatial phase convention. The electric field of such a plane wave is therefore

$$\vec{E} = \vec{E}_0 \psi = \vec{E}_0 e^{-i\vec{k}\cdot\vec{r}}, \tag{2.1.8}$$

where $\vec{E}_0$ is a constant phasor coefficient that sets the wave's reference field scale and polarization; any spatial attenuation remains in the exponential.
For the plane wave in (2.1.8), whose phasor coefficient $\vec{E}_0$ is constant, direct differentiation of its spatial factor gives

$$\begin{aligned} \vec{\nabla}\, e^{-i\vec{k}\cdot\vec{r}} &= \left(\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z\right) e^{-i(k_x x + k_y y + k_z z)} \\ &= \left(-ik_x\hat{x} + -ik_y\hat{y} + -ik_z\hat{z}\right) e^{-i\vec{k}\cdot\vec{r}}, \end{aligned}$$

simplifies to multiplication by $-i\vec{k}$:

$$\vec{\nabla} \Rightarrow -i\vec{k}. \tag{2.1.9}$$

The Plane Wave Solution (cont.)

In a charge-free, homogeneous, isotropic region with constant nonzero permittivity, Maxwell's equation $\vec{\nabla} \cdot \vec{D} = 0$ and $\vec{D} = \epsilon \vec{E}$ require $\vec{\nabla} \cdot \vec{E} = 0$. Applying this condition to the plane-wave solution gives

$$\vec{\nabla} \cdot \left(\vec{E}_0 e^{-i\vec{k}\cdot\vec{r}} \right) = -i \left(\vec{k} \cdot \vec{E}_0 \right) e^{-i\vec{k}\cdot\vec{r}} = 0. \tag{2.1.10}$$

For a lossless uniform plane wave, where the wave vector $\vec{k}$ is real, this result implies that $\vec{k} \cdot \vec{E}_0 = 0$, meaning the electric field is perpendicular to the direction of propagation. When $\vec{k}$ is complex, the relation $\vec{k} \cdot \vec{E}_0 = 0$ still holds; however, since this is a non-conjugated bilinear product, it no longer implies geometric perpendicularity of the *real* field vectors to the direction of propagation. A complex wavevector does not by itself mean that the wave is non-uniform: if its real and imaginary parts are parallel, the phase and amplitude planes are parallel and the result is an attenuated uniform plane wave; if those parts are not parallel, the wave is non-uniform (inhomogeneous). Both cases are discussed in Section 2.2.1. The special case in which the longitudinal component k_z is purely imaginary, while the medium wavenumber k remains real, is discussed in Section 2.1.2.

The Plane Wave Solution (cont.)

The magnetic field is then given by:

$$\begin{aligned}\vec{H} &= \frac{i}{\omega\mu}\vec{\nabla}\times\vec{E} = \frac{1}{\omega\mu}\vec{k}\times\vec{E} \\ &= \frac{1}{\eta}\hat{k}\times\vec{E} = \vec{H}_0 e^{-i\vec{k}\cdot\vec{r}},\end{aligned} \tag{2.1.11}$$

where $\vec{H}_0 = \frac{1}{\eta}\hat{k}\times\vec{E}_0$ and $\eta = \sqrt{\mu/\epsilon}$. For complex material parameters, the branches defining k and η are correlated so that $k/(\omega\mu) = 1/\eta$; $\hat{k}$ retains the algebraic meaning stated above. For a lossless uniform plane wave with real $\vec{k}$, these relations make $\vec{k}$, $\vec{E}$, and $\vec{H}$ mutually perpendicular, with $(\vec{E}, \vec{H}, \vec{k})$ forming a right-handed system. For a complex $\vec{k}$, the cross- and dot-product equations below remain valid as non-conjugated complex-vector identities, but they do not assert ordinary geometric orthogonality of instantaneous real field vectors. Substitution into the source-free Maxwell equations gives

$$\vec{k}\times\vec{E} = \omega\mu\vec{H}, \tag{2.1.12}$$

$$\vec{k}\times\vec{H} = -\omega\epsilon\vec{E}, \tag{2.1.13}$$

The Plane Wave Solution (cont.)

$$\vec{k} \cdot \vec{E} = 0, \tag{2.1.14}$$

$$\vec{k} \cdot \vec{H} = 0. \tag{2.1.15}$$

Notice that the above equations also apply to $(\vec{E}_0, \vec{H}_0)$.

The Angular Spectrum Representation

The angular spectrum representation expresses a sufficiently regular field on a transverse plane as a two-dimensional Fourier superposition. In a homogeneous, source-free half-space, each Fourier component continues away from that plane as a propagating or evanescent plane wave. For the real propagating/evanescent split used in this subsection, assume that the two half-spaces contain the same homogeneous, isotropic, lossless simple medium, that its wavenumber $k > 0$ is real, and that the Fourier variables k_x, k_y are real. The ordered comparisons $k_x^2 + k_y^2 \lesseqgtr k^2$ below then have their ordinary real meaning. In a lossy medium, one must instead use the appropriate passive complex branch of $k_z = \sqrt{k^2 - k_x^2 - k_y^2}$; the real inequalities and the real γ introduced below no longer provide the classification. In this subsection, we consider the problem of determining the outgoing electric field in the half spaces $z > 0$ and $z < 0$ from a prescribed tangential electric field distribution on the x–y plane. This is first a boundary-data construction, not by itself a specification of one physical current sheet. Indeed, let region 1 be the upper half-space, region 2 the lower half-space, and $\hat{n} = \hat{z}$. The construction below uses the same prescribed tangential field on the two sides, so (1.2.3) gives

$$\vec{M}_s = -\hat{z} \times (\vec{E}_1 - \vec{E}_2) = \vec{0},$$

The Angular Spectrum Representation (cont.)

whereas the oppositely directed outgoing waves generally have different tangential magnetic fields and therefore require

$$\vec{J}_s = \hat{z} \times (\vec{H}_1 - \vec{H}_2)$$

if they are to be interpreted as a two-sided field radiated by a sheet source. For example, at normal incidence in identical media, choosing $\vec{E}_1 = \vec{E}_2 = E_0\hat{x}$ gives $\vec{H}_1 = (E_0/\eta)\hat{y}$ and $\vec{H}_2 = -(E_0/\eta)\hat{y}$, hence $\vec{M}_s = \vec{0}$ and $\vec{J}_s = -(2E_0/\eta)\hat{x}$. In a *one-sided* surface-equivalence construction, the field on the suppressed side is instead set to a chosen value (often zero); the aperture field then contributes $\vec{M}_s = -\hat{n} \times (\vec{E}_1 - \vec{E}_2')$, generally alongside an equivalent electric current, with both signs fixed by the chosen normal. The goal is to represent the resulting field in terms of its angular spectrum components, which correspond to plane-wave contributions with different propagation directions. A similar approach will be applied later in Section 3.5 to analyze the radiation from an aperture. We expand an electromagnetic field by evaluating its two-dimensional inverse Fourier transform on a plane perpendicular to the z-axis:

$$\vec{\mathcal{E}}(k_x, k_y, z) = \frac{1}{(2\pi)^2} \iint_{-\infty}^{\infty} \vec{E}(x, y, z) e^{i(k_x x + k_y y)} dxdy, \qquad (2.1.16)$$

The Angular Spectrum Representation (cont.)

where x, y are the transverse position components in Cartesian coordinates and k_x, k_y are the corresponding spatial frequencies in k domain. The electromagnetic field itself can then be reconstructed by taking the two-dimensional Fourier transform of this spectral field:

$$\vec{E}(x, y, z) = \iint_{-\infty}^{\infty} \vec{\mathcal{E}}(k_x, k_y, z) e^{-i(k_x x + k_y y)} dk_x dk_y. \tag{2.1.17}$$

The transverse derivatives act only on the Fourier kernel:

$$\partial_x^2 \to (-ik_x)^2 = -k_x^2, \qquad \partial_y^2 \to (-ik_y)^2 = -k_y^2.$$

The z-derivatives instead pass through the integral and act on $\vec{\mathcal{E}}$. Inserting the reconstruction integral into the homogeneous vector Helmholtz equation $(\nabla^2 + k^2)\vec{E} = 0$ therefore gives

$$\iint_{-\infty}^{\infty} \left[\frac{d^2 \vec{\mathcal{E}}}{dz^2} + \left(k^2 - k_x^2 - k_y^2\right) \vec{\mathcal{E}} \right] e^{-i(k_x x + k_y y)} \, dk_x \, dk_y = 0.$$

Since the transverse exponentials are linearly independent, the bracketed quantity must vanish for every (k_x, k_y). Introducing $k_z^2 = k^2 - k_x^2 - k_y^2$, each Cartesian component

The Angular Spectrum Representation (cont.)

obeys a one-dimensional Helmholtz equation in z. For $k_z \neq 0$, its two independent solutions are $e^{\mp i k_z z}$. At the grazing point $k_z = 0$, the differential equation instead gives $A + Bz$; bounded outgoing data select the constant solution $B = 0$, which is also the $k_z \to 0$ limit used below. In the evanescent regime $k_x^2 + k_y^2 > k^2$, write $\gamma = \sqrt{k_x^2 + k_y^2 - k^2} > 0$; choosing $k_z = -i\gamma$ gives the decaying exponential $e^{-ik_z z} = e^{-\gamma z}$ in the half-space $z > 0$ and $e^{+ik_z z} = e^{\gamma z}$ in the half-space $z < 0$. With this branch convention,

$$k_z = \begin{cases} \sqrt{k^2 - k_x^2 - k_y^2}, & k_x^2 + k_y^2 \leq k^2, \\ -i\sqrt{k_x^2 + k_y^2 - k^2}, & k_x^2 + k_y^2 > k^2, \end{cases} \tag{2.1.18}$$

we can obtain the scalar Helmholtz equation for each Cartesian component of $\vec{\mathcal{E}}$, that is

$$\left(\frac{d^2}{dz^2} + k_z^2 \right) \mathcal{E}_\xi = 0, \tag{2.1.19}$$

where $\xi \in \{x, y, z\}$. Thus, we find that the spectral field propagates along the z-axis according to:

$$\vec{\mathcal{E}}_\pm(k_x, k_y, z) = \vec{\mathcal{E}}_\pm(k_x, k_y, 0) e^{\mp i k_z z}. \tag{2.1.20}$$

The Angular Spectrum Representation (cont.)

The upper sign and amplitude $\vec{\mathcal{E}}_+$ apply in $z > 0$; the lower sign and $\vec{\mathcal{E}}_-$ apply in $z < 0$. Their prescribed tangential components agree at the plane, but their normal components generally differ because the two wavevectors have opposite z components. When $k_x^2 + k_y^2 > k^2$, the upper sign gives the outgoing decaying solution for $z > 0$, while the lower sign gives the outgoing decaying solution for $z < 0$. A two-sided scalar component radiated by data on the plane can be written with the combined factor $e^{-\gamma|z|}$; a single nonzero exponential cannot remain finite as $z \to +\infty$ and as $z \to -\infty$.
Finally, the complete angular spectrum representation for the electric field is given by:

$$\vec{E}_\pm(x, y, z) = \iint_{-\infty}^{\infty} \vec{\mathcal{E}}_\pm(k_x, k_y, 0) e^{-i(k_x x + k_y y \pm k_z z)} dk_x dk_y. \tag{2.1.21}$$

While this representation satisfies the Helmholtz equation component by component, Maxwell's divergence equation also constrains each spectral amplitude. For the upper and lower half-spaces the corresponding wavevectors are

$$\vec{k}_\pm = k_x \hat{x} + k_y \hat{y} \pm k_z \hat{z},$$

The Angular Spectrum Representation (cont.)

and direct substitution gives

$$\vec{k}_\pm \cdot \vec{\mathcal{E}}_\pm = 0 \quad \Longrightarrow \quad \mathcal{E}_{z,\pm} = -\frac{k_x \mathcal{E}_{x,\pm} + k_y \mathcal{E}_{y,\pm}}{\pm k_z} \qquad (k_z \neq 0).$$

Thus, away from grazing, the prescribed two tangential components determine the longitudinal component; the magnetic spectrum then follows from $\vec{\mathcal{H}}_\pm = (\omega\mu)^{-1}\vec{k}_\pm \times \vec{\mathcal{E}}_\pm$. Exactly at $k_z = 0$, the divergence equation reduces to $k_x\mathcal{E}_x + k_y\mathcal{E}_y = 0$ and does not by itself determine $\mathcal{E}_z$; a grazing component must be specified separately or obtained as a bounded limit of nongrazing data.

A further analysis of the plane wave factor $e^{-i(k_x x + k_y y \pm k_z z)}$ gives two characteristic solutions:

$$\begin{aligned} & e^{-i(k_x x + k_y y \pm k_z z)}, && k_x^2 + k_y^2 \leq k^2, \\ & e^{-i(k_x x + k_y y)} \begin{cases} e^{-\gamma z}, & z > 0, \\ e^{\gamma z}, & z < 0, \end{cases} && k_x^2 + k_y^2 > k^2, \\ & && \gamma = \sqrt{k_x^2 + k_y^2 - k^2}. \end{aligned} \tag{2.1.22}$$

The first represents propagating waves. The second represents evanescent waves: they decay away from the plane and therefore do not deliver power to the far zone. An

The Angular Spectrum Representation (cont.)

individual evanescent component can nevertheless carry real power parallel to the plane and can store reactive energy in the normal direction; “evanescent” does not mean that its Poynting vector vanishes everywhere.

Polarization of Plane Waves

The polarization of an electromagnetic wave describes the orientation and time-variation of the electric field vector in the plane perpendicular to the direction of propagation. Consider a plane wave propagating in the $+z$-direction:

$$\vec{E} = (E_x\hat{x} + E_y\hat{y})e^{-ikz}, \tag{2.1.23}$$

where

$$\begin{cases} E_x = |E_x|e^{i\delta_x} \\ E_y = |E_y|e^{i\delta_y} \end{cases} \tag{2.1.24}$$

are the complex amplitudes of the x and y components of the electric field. We can classify the polarization of the wave into three main types:

Linear Polarization. A wave with both components nonzero is linearly polarized if their phase difference is an integer multiple of π:

$$\delta_x - \delta_y = n\pi, \quad n \in \mathbb{Z}. \tag{2.1.25}$$

In this case, the electric field vector oscillates along a fixed line in the $x - y$ plane. A wave with only one nonzero transverse component is also linearly polarized, although the phase of the zero component is undefined.

Polarization of Plane Waves (cont.)

Circular Polarization. A wave is circularly polarized if the amplitudes of its x and y components are equal and the phase difference is an odd multiple of $\pi/2$:

$$|E_x| = |E_y|, \quad \delta_x - \delta_y = \frac{(2m+1)\pi}{2}, \quad m \in \mathbb{Z}. \tag{2.1.26}$$

Handedness conventions differ between communities, so here we explicitly define right-hand circular polarization (RHCP) and left-hand circular polarization (LHCP) by

$$\vec{E}_{RC} = E_0\,(\hat{x} - i\hat{y})\,e^{-ikz}, \tag{2.1.27a}$$

$$\vec{E}_{LC} = E_0\,(\hat{x} + i\hat{y})\,e^{-ikz}, \tag{2.1.27b}$$

where E_0 is a real constant.
In order to characterize the rotation direction, by applying (1.4.4), we can transform (2.1.27) into time domain:

$$\vec{E}_{RC}(z,t) = E_0\left[\cos(\omega t - kz)\hat{x} + \sin(\omega t - kz)\hat{y}\right], \tag{2.1.28a}$$

$$\vec{E}_{LC}(z,t) = E_0\left[\cos(\omega t - kz)\hat{x} - \sin(\omega t - kz)\hat{y}\right]. \tag{2.1.28b}$$

Polarization of Plane Waves (cont.)

Setting $z = 0$, as t increases, $\vec{E}_{RC}$ rotates from $+\hat{x}$ toward $+\hat{y}$, whereas $\vec{E}_{LC}$ rotates from $+\hat{x}$ toward $-\hat{y}$. Thus, for an observer on the $+z$ axis looking toward the origin, the former appears counterclockwise and the latter clockwise. These equations, together with the stated viewing direction and the $e^{i\omega t}$ convention, remove the ambiguity; some optics references attach the opposite RHCP/LHCP names.

Elliptical Polarization. Elliptical polarization is the most general case, occurring when the conditions for linear or circular polarization are not met. In this case, the tip of the electric field vector traces out an ellipse in the plane perpendicular to the direction of propagation.

Many incoherent sources, including direct thermal emission, are approximately unpolarized: their polarization state fluctuates too rapidly to be represented by one deterministic polarization ellipse and must instead be described statistically. This is not universal. For example, scattering and reflection can make daylight partially polarized, so "unpolarized" describes the measured statistical state rather than every natural electromagnetic wave.

Phase and Group Velocities

For a plane wave propagating in the z-direction, its time-domain representation can be written as

$$\vec{E}(z,t) = \Re\left\{\vec{E}_0 e^{i(\omega t - kz)}\right\}. \tag{2.1.29}$$

In this discussion, assume a conventional lossless positive-index medium with real positive μ and ϵ, and choose the forward branch $k = \omega\sqrt{\mu\epsilon} > 0$. This assumption allows us to define the phase and group velocities in a straightforward manner. The planes of constant phase are defined by the condition

$$\omega t - kz = C, \tag{2.1.30}$$

where C is a constant. By taking the time derivative of this expression, we can determine the *phase velocity*, which is the speed at which these planes of constant phase propagate:

$$v_p = \frac{dz}{dt} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}}. \tag{2.1.31}$$

In a *non-dispersive* medium, where the permittivity ϵ and permeability μ do not vary with frequency, the phase velocity remains constant for all frequencies.

Phase and Group Velocities (cont.)

A simple harmonic wave of infinite duration has no modulation with which to encode new information. Modulation introduces a finite bandwidth, and closely spaced frequency components then form a wave packet. In a lossless medium whose dispersion is nearly linear across that narrow bandwidth, the envelope propagates at the *group velocity*. In that restricted regime the group velocity often approximates the energy-transport velocity. It is not, in general, the information-front velocity: strong dispersion, absorption, or pulse reshaping can make the group velocity superluminal, negative, or otherwise different from both energy and causal-signal velocities. To derive the narrow-band result, consider the following example.

For a narrow-band signal $s(t)$ modulated onto a high-frequency carrier $e^{i\omega_0 t}$, the overall envelope of the wave packet signal is $s(t)e^{i\omega_0 t}$. Let the Fourier transform of $s(t)$ be $S(\omega)$, then the Fourier transform of $s(t)e^{i\omega_0 t}$ will be $S(\omega - \omega_0)$. Suppose that the signal went through a distance z and was received by a receiver, then the field at the receiver can be expressed as

$$S_r(\omega, k) = S(\omega - \omega_0)e^{-ikz}. \tag{2.1.32}$$

Phase and Group Velocities (cont.)

Notice that the wavenumber k is also a function of ω. For narrow-band cases, we can expand k with the following expression

$$\begin{aligned} k &\approx k(\omega_0) + \left(dk/d\omega|_{\omega=\omega_0} \right) (\omega - \omega_0) \\ &= k_0 + \dot{k} \, (\omega - \omega_0) \, , \end{aligned} \tag{2.1.33}$$

where $k(\omega_0) = k_0$ and

$$\dot{k} = \left. \frac{dk}{d\omega} \right|_{\omega=\omega_0}$$

is the derivative of the wave number with respect to angular frequency, evaluated at the central frequency ω_0.

Plug the above equation into (2.1.32), we have

$$S_r(\omega, k) = S(\omega - \omega_0) e^{-ik_0 z} e^{-i(\omega - \omega_0)\dot{k}z}. \tag{2.1.34}$$

Phase and Group Velocities (cont.)

Inverse Fourier transform the above equation, we then get the time-domain signal at the receiver:

$$
\begin{aligned}
s_r(t) &= \mathfrak{F}^{-1}\{S_r(\omega)\} \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega-\omega_0)e^{-ik_0 z}e^{-i(\omega-\omega_0)\dot{k}z}e^{i\omega t}d\omega \\
&= \frac{e^{-ik_0 z}}{2\pi}\int_{-\infty}^{\infty} S(\Omega)e^{-i\dot{k}\Omega z}e^{i(\Omega+\omega_0)t}d\Omega \\
&= e^{i(\omega_0 t-k_0 z)}\,\frac{1}{2\pi}\int_{-\infty}^{\infty} S(\Omega)e^{i\Omega(t-\dot{k}z)}d\Omega \\
&= s(t-\dot{k}z)\,e^{i(\omega_0 t-k_0 z)},
\end{aligned}
\tag{2.1.35}
$$

Thus the envelope argument is constant when $t-\dot{k}z$ is constant. Differentiating $t-\dot{k}z=C$ gives $1-\dot{k}\,dz/dt=0$, so the envelope travels with group velocity

$$
v_g = \frac{1}{\dot{k}} = \frac{d\omega}{dk}. \tag{2.1.36}
$$

Phase and Group Velocities (cont.)

The relationship between the phase and group velocities is given by:

$$v_g = \frac{1}{dk/d\omega} = \frac{1}{d(\frac{\omega}{v_p})/d\omega} = \frac{v_p}{1 - \frac{\omega}{v_p}\frac{dv_p}{d\omega}}. \tag{2.1.37}$$

For $v_p > 0$, the sign of $dv_p/d\omega$ relates phase and group velocity through the denominator in (2.1.37). The comparisons below additionally assume that this denominator is positive; without that hypothesis v_g can diverge or become negative:

1. *No Dispersion*: If $dv_p/d\omega = 0$ throughout the signal bandwidth, then $v_g = v_p$ and those frequency components travel at the same speed.
2. *Normal Dispersion*: If $dv_p/d\omega < 0$, then $0 < v_g < v_p$. Equivalently, for a nonmagnetic transparent material with refractive index $n = c/v_p$, one has $dn/d\omega > 0$.
3. *Anomalous Dispersion*: If $dv_p/d\omega > 0$ and $1 - (\omega/v_p)(dv_p/d\omega) > 0$, then $v_g > v_p$. If that denominator is nonpositive, the simple inequality is replaced by an infinite or negative group velocity and pulse reshaping must be considered.

Phase and Group Velocities (cont.)

An illustration of normal dispersion is shown in Fig. 2.1. In most transparent dielectrics, the phase velocity v_p decreases with increasing angular frequency, such that $dv_p/d\omega < 0$. According to (2.1.37), the group velocity is therefore smaller than the phase velocity ($v_g < v_p$). Typical examples include visible light propagation in glass or water, where higher-frequency (shorter-wavelength) components travel more slowly than lower-frequency components.

When $dv_p/d\omega > 0$ and the denominator of (2.1.37) remains positive, the group velocity is greater than the phase velocity. Anomalous dispersion commonly occurs in frequency intervals near material resonances; it is not restricted to metamaterials and is often accompanied by appreciable absorption, where a real-k envelope argument alone is insufficient.

Phase and Group Velocities

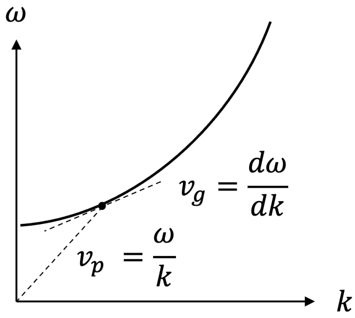

Figure 2.1: On an ω–k dispersion curve, $v_p = \omega/k$ is the slope of the line from the origin and $v_g = d\omega/dk$ is the local tangent slope. The curve shown has $v_g < v_p$ at the marked point.

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Propagation in an Unbounded Medium

In a lossless simple medium with real positive μ and ϵ, the scalar wavenumber $k = \omega\sqrt{\mu\epsilon}$ is real. The normalized direction vector $\hat{k}$ can nevertheless be complex for an inhomogeneous plane wave. Write

$$\hat{k} = \vec{k}' - i\vec{k}'', \tag{2.2.1}$$

where $\vec{k}'$ and $\vec{k}''$ are real vectors, then the condition $\hat{k} \cdot \hat{k} = 1$ implies:

$$(k')^2 - (k'')^2 - 2i\vec{k}' \cdot \vec{k}'' = 1, \tag{2.2.2}$$

where $k' = |\vec{k}'|$ and $k'' = |\vec{k}''|$. This single complex equation is equivalent to two real equations:

$$(k')^2 - (k'')^2 = 1, \tag{2.2.3a}$$

$$\vec{k}' \cdot \vec{k}'' = 0. \tag{2.2.3b}$$

The simplest case is that of a uniform plane wave, where

$$k' = 1, \quad k'' = 0. \tag{2.2.4}$$

Propagation in an Unbounded Medium (cont.)

In the general case, where both $\vec{k}'$ and $\vec{k}''$ are non-zero, the solution $\vec{E} = \vec{E}_0 e^{-i\vec{k}\cdot\vec{r}}$ takes the form:

$$\vec{E} = \vec{E}_0 e^{-k\,\vec{k}''\cdot\vec{r}} e^{-ik\,\vec{k}'\cdot\vec{r}}. \tag{2.2.5}$$

This represents a non-uniform plane wave, in which the planes of constant phase are perpendicular to the real part of the wavevector, $\vec{k}'$, and the planes of constant amplitude are perpendicular to the imaginary part, $\vec{k}''$. According to (2.2.3b), $\vec{k}'$ and $\vec{k}''$ are orthogonal.

In a passive lossy simple medium, assume here that $\mu > 0$ is real, $\epsilon_c = \epsilon' - i\epsilon''$ with $\epsilon' > 0$, $\epsilon'' \geq 0$, and the square-root branch is chosen so that

$$k = \omega\sqrt{\mu\epsilon_c} = \beta - i\alpha, \qquad \beta \geq 0, \quad \alpha \geq 0.$$

If the direction vector $\hat{k}$ is real, the solution takes the form

$$\vec{E} = \vec{E}_0 e^{-i\vec{k}\cdot\vec{r}} = \vec{E}_0 e^{-\alpha\hat{k}\cdot\vec{r}} e^{-i\beta\hat{k}\cdot\vec{r}}. \tag{2.2.6}$$

This represents an attenuated uniform plane wave, not an evanescent wave: the constant-phase and constant-amplitude planes are both normal to $\hat{k}$, and the

Propagation in an Unbounded Medium (cont.)

time-averaged power still flows along $\hat{k}$ while decaying with distance. To obtain the propagation constant β and attenuation constant α, square both representations of k:

$$(\beta - i\alpha)^2 = \beta^2 - \alpha^2 - 2i\alpha\beta = \omega^2\mu(\epsilon' - i\epsilon'').$$

Equating real and imaginary parts, and then adding the squares of the two equations, gives

$$\begin{aligned} \beta^2 - \alpha^2 &= \omega^2\mu\epsilon', \\ 2\alpha\beta &= \omega^2\mu\epsilon'', \\ (\beta^2 + \alpha^2)^2 &= (\beta^2 - \alpha^2)^2 + (2\alpha\beta)^2 \\ &= \omega^4\mu^2\left[(\epsilon')^2 + (\epsilon'')^2\right]. \end{aligned}$$

Because $\alpha, \beta \geq 0$,

$$\beta^2 + \alpha^2 = \omega^2\mu\epsilon'\sqrt{1 + (\epsilon''/\epsilon')^2} = \omega^2\mu\epsilon'\sqrt{1 + \tan^2\delta}.$$

Propagation in an Unbounded Medium (cont.)

Adding and subtracting this equation and $\beta^2 - \alpha^2 = \omega^2\mu\epsilon'$ yields

$$\beta = \omega\left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+(\tan\delta)^2}+1\right]\right\}^{1/2}, \tag{2.2.7}$$

$$\alpha = \omega\left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+(\tan\delta)^2}-1\right]\right\}^{1/2}. \tag{2.2.8}$$

The intrinsic impedance and phase velocity are given by

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}}\,(1 - i\tan\delta)^{-1/2}, \tag{2.2.9}$$

$$v_p = \frac{\omega}{\beta} = \left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+(\tan\delta)^2}+1\right]\right\}^{-1/2}. \tag{2.2.10}$$

Oblique Incidence at an Interface

We now examine a plane wave at a planar, source-free boundary between two homogeneous, isotropic simple media. The Fresnel derivation permits complex material parameters and angles when passive, mutually correlated square-root branches are used: $k_n/(\omega\mu_n) = 1/\eta_n$ must hold, and the transmitted normal wavenumber $k_{2x} = k_2 \cos\theta_t$ is selected so that $e^{-ik_{2x}x}$ is outgoing and non-growing in $x > 0$. Under the present $e^{i\omega t}$ convention, a decaying transmitted branch has $\Im k_{2x} < 0$; when k_{2x} is real, its sign is fixed by outward energy flow. The later discussions of real critical and Brewster angles will explicitly restrict both media to lossless, positive-parameter media. Special cases include normal incidence ($\theta_i = 0$) and a PEC limit, obtained for fixed μ_2 by taking a passive conducting permittivity $\epsilon_2 \to -i\infty$ under the present $e^{i\omega t}$ convention.
Let the incident electric-field basis coefficient be E_i. Define the *reflection* and *transmission coefficients*, Γ and T, as the corresponding reflected and transmitted phasor-coefficient ratios:

$$\Gamma = \frac{E_r}{E_i}, \tag{2.2.11}$$

$$T = \frac{E_t}{E_i}, \tag{2.2.12}$$

Oblique Incidence at an Interface (cont.)

For propagating waves at real angles, these coefficients are also ratios of total electric-field amplitudes. For an evanescent or lossy transmitted wave, the polarization basis itself is complex, so T should be interpreted as a basis coefficient rather than as a Euclidean vector magnitude. We consider the two principal polarizations, *perpendicular* and *parallel*, illustrated in Fig. 2.2; any incident polarization can be decomposed into these two independent components.

Oblique Incidence at an Interface

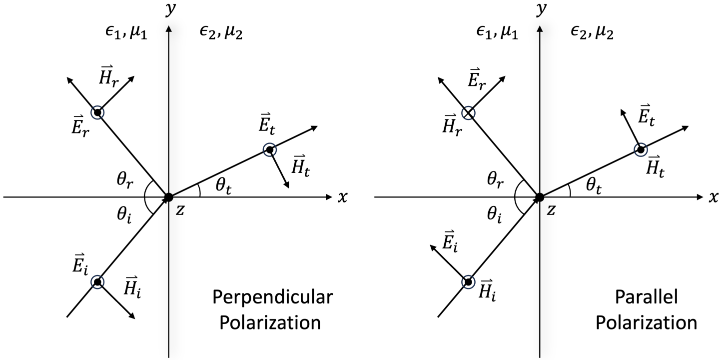

Figure 2.2: Oblique incidence at the plane $x = 0$. A dot (cross) denotes a field component along $+\hat{z}$ $(-\hat{z})$; the field arrows agree with the bases used in the perpendicular- and parallel-polarization equations.

Oblique Incidence at an Interface

Perpendicular Polarization. For perpendicular polarization, the incident electric field is normal to the plane of incidence. Take the interface to be $x = 0$, with $\hat{x}$ directed from medium 1 into medium 2. For positive propagating incidence, the principal incident and transmitted angles $0 \leq \theta_i, \theta_t \leq \pi/2$ are measured from $+\hat{x}$, while $0 \leq \theta_r \leq \pi/2$ is measured from $-\hat{x}$, as shown in Fig. 2.2. The transmitted angle is analytically continued onto the passive complex branch in the total-internal-reflection discussion below. Write $\vec{k}_i = k_1 \hat{k}_i$ and, for medium $n = 1, 2$, define $k_n = \omega\sqrt{\epsilon_n \mu_n}$ and $\eta_n = \sqrt{\mu_n/\epsilon_n}$. From (2.1.8) and (2.1.12),

$$\vec{E}_i = E_i e^{-i\vec{k}_i \cdot \vec{r}} = E_i e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}\hat{z}, \tag{2.2.13a}$$

$$\begin{aligned} \vec{H}_i &= \frac{1}{\eta_1}\hat{k}_i \times \vec{E}_i = \frac{1}{\eta_1}(\cos\theta_i \hat{x} + \sin\theta_i \hat{y}) \times \vec{E}_i \\ &= \frac{E_i}{\eta_1} e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}(\sin\theta_i \hat{x} - \cos\theta_i \hat{y}), \end{aligned} \tag{2.2.13b}$$

The reflected and transmitted fields are expressed similarly:

$$\vec{E}_r = \Gamma_\perp E_i e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)}\hat{z}, \tag{2.2.14a}$$

Oblique Incidence at an Interface (cont.)

$$\vec{H}_r = \frac{\Gamma_\perp E_i}{\eta_1} e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)}(\sin\theta_r\hat{x} + \cos\theta_r\hat{y}), \tag{2.2.14b}$$

$$\vec{E}_t = T_\perp E_i e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}\hat{z}, \tag{2.2.15a}$$

$$\vec{H}_t = \frac{T_\perp E_i}{\eta_2} e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}(\sin\theta_t\hat{x} - \cos\theta_t\hat{y}). \tag{2.2.15b}$$

Applying the matching conditions for the tangential components of the electric and magnetic fields at the interface ($x = 0$):

$$(\vec{E}_i + \vec{E}_r)|^z_{x=0} = \vec{E}_t|^z_{x=0}, \tag{2.2.16a}$$

$$(\vec{H}_i + \vec{H}_r)|^y_{x=0} = \vec{H}_t|^y_{x=0}, \tag{2.2.16b}$$

we get

$$e^{-ik_1 y\sin\theta_i} + \Gamma_\perp e^{-ik_1 y\sin\theta_r} = T_\perp e^{-ik_2 y\sin\theta_t}, \tag{2.2.17a}$$

$$\frac{1}{\eta_1}\left(-\cos\theta_i e^{-ik_1 y\sin\theta_i} + \Gamma_\perp\cos\theta_r e^{-ik_1 y\sin\theta_r}\right) = -\frac{1}{\eta_2}T_\perp\cos\theta_t e^{-ik_2 y\sin\theta_t}. \tag{2.2.17b}$$

Oblique Incidence at an Interface (cont.)

For these boundary conditions to hold for all y on the interface, the phase terms must be equal. This requirement leads to the phase-matching condition, also known as *Snell's law*:

$$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t. \tag{2.2.18}$$

This result implies that the angle of reflection equals the angle of incidence, $\theta_r = \theta_i$. Using the phase-matching condition, the boundary condition equations simplify to:

$$1 + \Gamma_\perp = T_\perp, \tag{2.2.19a}$$

$$\frac{\cos\theta_i}{\eta_1}(-1 + \Gamma_\perp) = -\frac{\cos\theta_t}{\eta_2} T_\perp. \tag{2.2.19b}$$

Solving (2.2.19), we can derive the reflection and transmission coefficients for perpendicular polarization:

$$\frac{\cos\theta_i}{\eta_1}\left(\Gamma_\perp - 1\right) = -\frac{\cos\theta_t}{\eta_2}\left(1 + \Gamma_\perp\right),$$

Oblique Incidence at an Interface (cont.)

where we have used (2.2.19) to eliminate $T_\perp = 1 + \Gamma_\perp$. Collecting the terms proportional to $\Gamma_\perp$ on one side gives

$$\frac{\cos\theta_i}{\eta_1} - \frac{\cos\theta_t}{\eta_2} = \Gamma_\perp \left(\frac{\cos\theta_i}{\eta_1} + \frac{\cos\theta_t}{\eta_2} \right),$$

so that

$$\Gamma_\perp = \frac{\cos\theta_i/\eta_1 - \cos\theta_t/\eta_2}{\cos\theta_i/\eta_1 + \cos\theta_t/\eta_2}.$$

Multiplying numerator and denominator by $\eta_1\eta_2/(\cos\theta_i \cos\theta_t)$ recasts $\Gamma_\perp$ in terms of the tangential wave impedances, and substituting the result back into $T_\perp = 1 + \Gamma_\perp$ then yields:

$$\Gamma_\perp = \frac{\eta_2/\cos\theta_t - \eta_1/\cos\theta_i}{\eta_2/\cos\theta_t + \eta_1/\cos\theta_i}, \tag{2.2.20a}$$

$$T_\perp = \frac{2\eta_2/\cos\theta_t}{\eta_2/\cos\theta_t + \eta_1/\cos\theta_i}. \tag{2.2.20b}$$

The terms $\eta_1/\cos\theta_i$ and $\eta_2/\cos\theta_t$ can be interpreted as the wave impedances for the tangential field components.

Oblique Incidence at an Interface (cont.)

Parallel Polarization. For parallel polarization, the incident electric field lies within the plane of incidence. Following similar derivations, the corresponding fields are expressed as:

$$\vec{E}_i = E_i e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}(-\sin\theta_i \hat{x} + \cos\theta_i \hat{y}), \tag{2.2.21a}$$

$$\vec{H}_i = \frac{E_i}{\eta_1} e^{-ik_1(x\cos\theta_i + y\sin\theta_i)} \hat{z}, \tag{2.2.21b}$$

$$\vec{E}_r = \Gamma_\parallel E_i e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)}(\sin\theta_r \hat{x} + \cos\theta_r \hat{y}), \tag{2.2.22a}$$

$$\vec{H}_r = -\frac{\Gamma_\parallel E_i}{\eta_1} e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)} \hat{z}, \tag{2.2.22b}$$

$$\vec{E}_t = T_\parallel E_i e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}(-\sin\theta_t \hat{x} + \cos\theta_t \hat{y}), \tag{2.2.23a}$$

$$\vec{H}_t = \frac{T_\parallel E_i}{\eta_2} e^{-ik_2(x\cos\theta_t + y\sin\theta_t)} \hat{z}. \tag{2.2.23b}$$

Oblique Incidence at an Interface (cont.)

At $x = 0$, phase matching again gives $\theta_r = \theta_i$. Continuity of the tangential components E_y and H_z then gives, without skipping the projection factors,

$$\cos\theta_i \left(1 + \Gamma_\parallel\right) = T_\parallel \cos\theta_t, \tag{2.2.24a}$$

$$\frac{1 - \Gamma_\parallel}{\eta_1} = \frac{T_\parallel}{\eta_2}. \tag{2.2.24b}$$

Eliminating $T_\parallel$ yields

$$\eta_1 \cos\theta_i \left(1 + \Gamma_\parallel\right) = \eta_2 \cos\theta_t \left(1 - \Gamma_\parallel\right),$$

and hence

$$\Gamma_\parallel = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}, \tag{2.2.25a}$$

$$T_\parallel = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}. \tag{2.2.25b}$$

For real nongrazing angles, this result may equivalently be displayed as a tangential-field coefficient times a projection factor,

$$T_\parallel = \frac{E_t}{E_i} = \frac{E_{yt}/\cos\theta_t}{E_{yi}/\cos\theta_i} = \frac{2\eta_2 \cos\theta_t}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i} \frac{\cos\theta_i}{\cos\theta_t}.$$

Oblique Incidence at an Interface (cont.)

The first form in (2.2.25b) remains finite in the grazing-transmission limit $\cos\theta_t \to 0$ provided $\cos\theta_i \neq 0$, whereas the factored form is only a nongrazing algebraic rewriting. If both cosines vanish simultaneously, the coefficient is defined by its one-sided Fresnel limit, not by substituting $0/0$. For lossless media with $\epsilon_n > 0$, $\mu_n > 0$, and real transmission angle, the amplitude coefficients convert to normal power-flux coefficients as follows. A plane wave of total electric-field amplitude E has normal time-averaged flux $|E|^2 \cos\theta/(2\eta)$, so

$$\mathcal{R} = |\Gamma|^2, \qquad \mathcal{T} = \frac{\eta_1}{\eta_2}\frac{\cos\theta_t}{\cos\theta_i}|T|^2, \qquad \mathcal{R} + \mathcal{T} = 1. \tag{2.2.26}$$

The last equality follows by substituting either polarization's Fresnel coefficients and using Snell's law; it expresses conservation of normal time-averaged power. These power formulas should not be applied by simply inserting a complex θ_t in the total-internal-reflection regime; that case is evaluated directly from $\frac{1}{2}\Re\{\vec{E} \times \vec{H}^*\}$ below.

The Critical Angle and Total Internal Reflection. For this paragraph assume lossless media with real positive ϵ_n and μ_n. Total internal reflection can occur when the incident

Oblique Incidence at an Interface (cont.)

medium has the larger wavenumber, $k_1 > k_2$, equivalently $\sqrt{\epsilon_1\mu_1} > \sqrt{\epsilon_2\mu_2}$. The critical angle θ_c is the grazing-transmission limit $\theta_t = 90^\circ$. From (2.2.18),

$$\sqrt{\epsilon_1\mu_1}\sin\theta_c = \sqrt{\epsilon_2\mu_2}, \tag{2.2.27}$$

or,

$$\theta_c = \sin^{-1}\left(\sqrt{\frac{\epsilon_2\mu_2}{\epsilon_1\mu_1}}\right). \tag{2.2.28}$$

For incidence above the critical angle, $\theta_i > \theta_c$, the wave is totally reflected and the transmitted field is evanescent; at $\theta_i = \theta_c$ the transmitted wave is the grazing limiting case. To understand what happens to the transmitted field, let us examine the case of perpendicular polarization. The transmitted field is given by

$$\begin{aligned} E_t &= T_\perp E_i e^{-ik_2(x\cos\theta_t + y\sin\theta_t)} \\ &= T_\perp E_i e^{-ik_1 y\sin\theta_i} e^{-ik_2 x\cos\theta_t}, \end{aligned} \tag{2.2.29}$$

where

$$\cos\theta_t = \sqrt{1-\sin^2\theta_t} = \sqrt{1-\frac{\epsilon_1\mu_1}{\epsilon_2\mu_2}\sin^2\theta_i}. \tag{2.2.30}$$

Oblique Incidence at an Interface (cont.)

When $\theta_i > \theta_c$, the term under the square root in (2.2.30) becomes negative, making $\cos\theta_t$ purely imaginary, as illustrated in Fig. 2.3. This figure shows that when the incident angle exceeds the critical angle, the refracted ray no longer propagates into the second medium but instead gives rise to an evanescent field that decays exponentially away from the interface.

The complex transmission angle can be expressed as

$$\begin{aligned}\cos\theta_t &= \cos\left(\frac{\pi}{2} + i\delta_t\right) = -i\sinh\delta_t \\ &= -i\sqrt{\frac{\epsilon_1\mu_1}{\epsilon_2\mu_2}\sin^2\theta_i - 1},\end{aligned} \tag{2.2.31}$$

where $\sinh\delta_t > 0$. The parameter δ_t is dimensionless; the actual normal amplitude-attenuation constant is $k_2\sinh\delta_t$, with units m^{-1}. Physically, $\theta_t = \frac{\pi}{2} + i\delta_t$ means that the real phase vector is parallel to the interface, so the constant-phase fronts are perpendicular to the interface, while the field amplitude decreases in the normal direction.

Oblique Incidence at an Interface (cont.)

Substituting this expression into (2.2.29), the transmitted field becomes

$$E_t = T_\perp E_i e^{-\sinh \delta_t k_2 x} e^{-ik_1 y \sin \theta_i}. \tag{2.2.32}$$

This represents an evanescent field confined near the interface, with its amplitude decaying into the optically rarer medium.
This is the mathematical form of a non-uniform plane wave, as seen in (2.2.5). It is important to distinguish phase propagation from normal power transport. For example, for the perpendicular field above, $\sin\theta_t$ is real and $\cos\theta_t = -i\sinh\delta_t$, and direct use of the transmitted $\vec{E}_t, \vec{H}_t$ gives

$$\left\langle \vec{S}_t \right\rangle \cdot \hat{x} = \frac{1}{2}\Re\left\{ (\vec{E}_t \times \vec{H}_t^*) \cdot \hat{x} \right\} = 0,$$

whereas its component parallel to the interface is generally nonzero. Thus no time-averaged power crosses into the lossless second half-space, even though a near field exists there. Correspondingly, substitution of $\cos\theta_t = -i\sinh\delta_t$ into either Fresnel reflection coefficient gives $|\Gamma| = 1$.

Oblique Incidence at an Interface

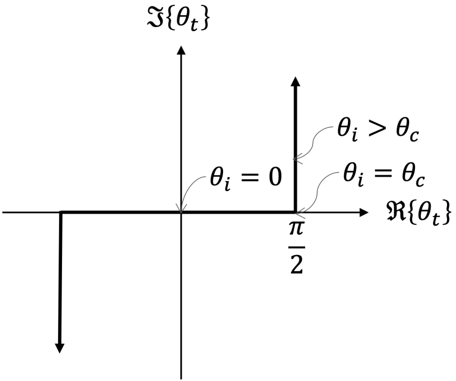

Figure 2.3: Complex transmission-angle plane. For positive incidence, $0 \leq \theta_i \leq \theta_c$ maps along the real segment $0 \leq \theta_t \leq \pi/2$, while $\theta_i > \theta_c$ follows the physical branch $\theta_t = \pi/2 + i\delta_t$ upward. The downward branch at $-\pi/2$ is the symmetric continuation for negative incidence.

Oblique Incidence at an Interface

Brewster's Angle. Continue to assume lossless media with $\epsilon_n > 0$ and $\mu_n > 0$. Here a Brewster angle θ_b is a nongrazing real incidence angle, $0 \leq \theta_b < \pi/2$, for which the relevant reflection coefficient is zero. The value at $\theta_i = \pi/2$ must instead be understood as a one-sided Fresnel limit; the common vanishing or diverging projection factors at exact grazing cannot be cancelled before that limit is taken. Define the positive material ratios

$$e = \frac{\epsilon_2}{\epsilon_1}, \qquad m = \frac{\mu_2}{\mu_1}, \qquad s = \sin^2\theta_b.$$

Snell's law then gives $\sin^2\theta_t = s/(em)$.
For perpendicular polarization, $\Gamma_\perp = 0$ requires $\eta_2/\cos\theta_t = \eta_1/\cos\theta_b$. Squaring this equation and using $\eta_2^2/\eta_1^2 = m/e$ gives

$$\frac{m/e}{1 - s/(em)} = \frac{1}{1-s} \quad \Longrightarrow \quad s\left(\frac{m}{e} - \frac{1}{em}\right) = \frac{m}{e} - 1.$$

When $m \neq 1$, solving for s yields

$$\sin\theta_b = \sqrt{\frac{e-m}{1/m - m}} = \sqrt{\frac{\epsilon_2/\epsilon_1 - \mu_2/\mu_1}{\mu_1/\mu_2 - \mu_2/\mu_1}}. \tag{2.2.33}$$

Oblique Incidence at an Interface (cont.)

A real nongrazing angle exists precisely when the radicand lies in $[0, 1)$ (and an interior Brewster angle has radicand in $(0, 1)$). An absolute-value inequality alone is not sufficient, because it would incorrectly admit a negative radicand. If $m = 1$ but $e \neq 1$, no perpendicular-polarization Brewster angle exists. If $m = e = 1$, the media are identical and $\Gamma_\perp = 0$ at every nongrazing angle rather than at one distinguished angle. For parallel polarization, $\Gamma_\| = 0$ requires $\eta_2 \cos\theta_t = \eta_1 \cos\theta_b$. Squaring and substituting Snell's law gives

$$\frac{m}{e}\left(1 - \frac{s}{em}\right) = 1 - s \quad \Longrightarrow \quad s\left(1 - \frac{1}{e^2}\right) = 1 - \frac{m}{e}.$$

For $e \neq 1$ this gives

$$\sin\theta_b = \sqrt{\frac{e - m}{e - 1/e}} = \sqrt{\frac{\epsilon_2/\epsilon_1 - \mu_2/\mu_1}{\epsilon_2/\epsilon_1 - \epsilon_1/\epsilon_2}}. \tag{2.2.34}$$

Again, a real nongrazing angle exists precisely when the radicand is in $[0, 1)$. If $e = 1$ but $m \neq 1$, there is no parallel-polarization Brewster angle; if $e = m = 1$, both polarizations are reflectionless at every nongrazing angle.

Oblique Incidence at an Interface (cont.)

The excluded algebraic endpoint $s = 1$ is not a Brewster root. In either of the nonexceptional formulas above, setting $s = 1$ forces $em = 1$. Snell's law then gives $\theta_t = \theta_i$ for every nongrazing angle, and both Fresnel coefficients reduce before taking the grazing limit to

$$\Gamma_\perp = \Gamma_\parallel = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}.$$

This limit is nonzero unless $\eta_2 = \eta_1$; together with $em = 1$, impedance equality implies $e = m = 1$, the identical-media case already handled separately. Thus admitting a radicand equal to one would retain a spurious solution introduced when the squared zero-reflection equations are evaluated after both angle cosines have vanished.
For the common nonmagnetic case $m = 1$ with positive permittivities, the parallel-polarization result reduces (for unequal media) to

$$\begin{aligned} \theta_b &= \sin^{-1}\left(\sqrt{\frac{\epsilon_2}{\epsilon_1 + \epsilon_2}}\right) = \cos^{-1}\left(\sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}}\right) \\ &= \tan^{-1}\left(\sqrt{\frac{\epsilon_2}{\epsilon_1}}\right), \end{aligned} \tag{2.2.35}$$

which is the usual Brewster angle.

Oblique Incidence at an Interface (cont.)

It is worth noting that since $\sqrt{\epsilon_2}\sin\theta_t = \sqrt{\epsilon_1}\sin\theta_b = \sqrt{\frac{\epsilon_1\epsilon_2}{\epsilon_1+\epsilon_2}}$, we have

$$\begin{aligned}\theta_t &= \sin^{-1}\left(\sqrt{\frac{\epsilon_1}{\epsilon_1+\epsilon_2}}\right) = \cos^{-1}\left(\sqrt{\frac{\epsilon_2}{\epsilon_1+\epsilon_2}}\right) \\ &= \tan^{-1}\left(\sqrt{\frac{\epsilon_1}{\epsilon_2}}\right).\end{aligned} \tag{2.2.36}$$

Thus, we have the following relationship:

$$\theta_b + \theta_t = \frac{\pi}{2}. \tag{2.2.37}$$

The relationship between θ_b and θ_t can be visualized, as shown in Fig. 2.4.

Oblique Incidence at an Interface

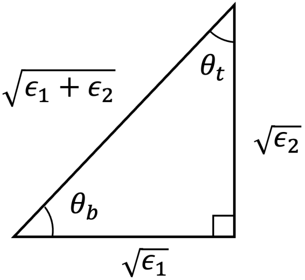

Figure 2.4: For positive nonmagnetic dielectric media, the side lengths $\sqrt{\epsilon_1}$ and $\sqrt{\epsilon_2}$ give $\tan\theta_b = \sqrt{\epsilon_2/\epsilon_1}$ and $\tan\theta_t = \sqrt{\epsilon_1/\epsilon_2}$, hence $\theta_b + \theta_t = \pi/2$.

Oblique Incidence at an Interface

For parallel polarization at a nonmagnetic dielectric interface, a useful microscopic picture is that the transmitted electric field drives polarization dipoles in medium 2. An ideal electric dipole does not radiate along its oscillation axis, as discussed in Section 3.3. At the Brewster condition $\theta_b + \theta_t = \pi/2$, that axis is parallel to the would-be reflected ray, so the reflected far-field contribution vanishes. This is a heuristic explanation of the nonmagnetic parallel-polarization result; the Fresnel boundary-condition derivation above is the general proof and also covers magnetic media.

Table of contents

Solving Waveguide Problems

Waveguides are structures designed to guide electromagnetic waves. Depending on conductor topology they may support TEM modes, and they can also support TE and TM modes. In this section the guide is uniform along z and its filling medium is homogeneous and isotropic. We first determine the longitudinal field components (E_z and H_z) and then reconstruct the transverse components. Assume

$$\vec{E}(x,y,z) = \vec{E}_0(x,y)e^{-ik_z z}, \tag{2.3.1a}$$

$$\vec{H}(x,y,z) = \vec{H}_0(x,y)e^{-ik_z z}. \tag{2.3.1b}$$

By separating the fields into their transverse and longitudinal components (e.g., $\vec{E}_0 = \vec{E}^0_\perp + E^0_z\hat{z}$) and substituting these into the source-free Maxwell equations with $\partial_z \to -ik_z$, we can express the transverse field components in terms of the longitudinal components E^0_z and H^0_z: With $\partial_z \to -ik_z$, the del operator splits as $\vec{\nabla} = \vec{\nabla}_\perp - ik_z\hat{z}$, so the source-free curl equations $\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H}$ and $\vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E}$ become $\left(\vec{\nabla}_\perp - ik_z\hat{z}\right) \times \left(\vec{E}^0_\perp + E^0_z\hat{z}\right) = -i\omega\mu\left(\vec{H}^0_\perp + H^0_z\hat{z}\right)$ together with its dual. Projecting each onto the transverse plane, and noting that $\vec{\nabla}_\perp \times \vec{E}^0_\perp$ is purely longitudinal while $\hat{z} \times (E^0_z\hat{z}) = \vec{0}$, the transverse parts read

$$-\hat{z} \times \vec{\nabla}_\perp E^0_z - ik_z\,\hat{z} \times \vec{E}^0_\perp = -i\omega\mu\,\vec{H}^0_\perp,$$

Solving Waveguide Problems (cont.)

$$-\hat{z} \times \vec{\nabla}_\perp H_z^0 - ik_z \, \hat{z} \times \vec{H}_\perp^0 = \quad i\omega\epsilon \, \vec{E}_\perp^0,$$

where we used $\vec{\nabla}_\perp E_z^0 \times \hat{z} = -\hat{z} \times \vec{\nabla}_\perp E_z^0$. Crossing the first with $\hat{z}$ and applying $\hat{z} \times (\hat{z} \times \vec{V}_\perp) = -\vec{V}_\perp$ gives

$$\hat{z} \times \vec{H}_\perp^0 = \frac{i}{\omega\mu} \left(\vec{\nabla}_\perp E_z^0 + ik_z \vec{E}_\perp^0 \right),$$

which, inserted into the second transverse equation, eliminates $\vec{H}_\perp^0$:

$$i\omega\epsilon \, \vec{E}_\perp^0 = -\hat{z} \times \vec{\nabla}_\perp H_z^0 + \frac{k_z}{\omega\mu} \left(\vec{\nabla}_\perp E_z^0 + ik_z \vec{E}_\perp^0 \right).$$

Collecting the $\vec{E}_\perp^0$ terms and using $k_c^2 = \omega^2\mu\epsilon - k_z^2$ clears the denominator. To display the dual elimination as well, crossing the second transverse equation with $\hat{z}$ gives

$$\hat{z} \times \vec{E}_\perp^0 = -\frac{i}{\omega\epsilon} \left(\vec{\nabla}_\perp H_z^0 + ik_z \vec{H}_\perp^0 \right).$$

Solving Waveguide Problems (cont.)

Substitution into the first transverse equation gives

$$-\hat{z}\times\vec{\nabla}_\perp E_z^0 - \frac{k_z}{\omega\epsilon}\vec{\nabla}_\perp H_z^0 = -\frac{i(\omega^2\mu\epsilon - k_z^2)}{\omega\epsilon}\vec{H}_\perp^0,$$

and hence the two reconstructed transverse fields are

$$\vec{E}_\perp^0 = -\frac{i}{k_c^2}\left(k_z\vec{\nabla}_\perp E_z^0 - \omega\mu\,\hat{z}\times\vec{\nabla}_\perp H_z^0\right), \tag{2.3.2a}$$

$$\vec{H}_\perp^0 = -\frac{i}{k_c^2}\left(k_z\vec{\nabla}_\perp H_z^0 + \omega\epsilon\,\hat{z}\times\vec{\nabla}_\perp E_z^0\right). \tag{2.3.2b}$$

where $\vec{\nabla}_\perp = (\hat{x}\partial_x + \hat{y}\partial_y)$ and

$$k_c^2 = k^2 - k_z^2. \tag{2.3.3}$$

Equations (2.3.2) require $k_c \neq 0$. A TEM mode has $E_z^0 = H_z^0 = 0$ and $k_c = 0$, so its nonzero transverse fields cannot be recovered by dividing these longitudinal fields by k_c^2; TEM modes must be solved directly from the transverse electrostatic potential and require a multiply connected conductor geometry. A hollow guide bounded by one connected conductor has no nontrivial TEM mode, as addressed in Problem 6.
These general expressions can be simplified for specific mode types:

Solving Waveguide Problems (cont.)

TM Modes. For Transverse Magnetic (TM) modes, the magnetic field is purely transverse to the direction of propagation, so $H_z^0 = 0$ and $E_z^0 \neq 0$. The transverse fields are then given by:

$$H_z^0 = 0, \tag{2.3.4a}$$

$$\vec{E}_\perp^0 = -\frac{i}{k_c^2} k_z \vec{\nabla}_\perp E_z^0, \tag{2.3.4b}$$

$$\vec{H}_\perp^0 = -\frac{i}{k_c^2} \omega\epsilon\, \hat{z} \times \vec{\nabla}_\perp E_z^0. \tag{2.3.4c}$$

TE Modes. For Transverse Electric (TE) modes, the electric field is purely transverse, so $E_z^0 = 0$ and $H_z^0 \neq 0$. The transverse fields are then:

$$E_z^0 = 0, \tag{2.3.5a}$$

$$\vec{E}_\perp^0 = \frac{i}{k_c^2} \omega\mu\, \hat{z} \times \vec{\nabla}_\perp H_z^0, \tag{2.3.5b}$$

$$\vec{H}_\perp^0 = -\frac{i}{k_c^2} k_z \vec{\nabla}_\perp H_z^0. \tag{2.3.5c}$$

Solving Waveguide Problems (cont.)

Each Cartesian component of the source-free fields satisfies the scalar Helmholtz equation. For either longitudinal amplitude $\psi \in \{E_z^0, H_z^0\}$, substitution of the separated z dependence in (2.3.1) gives, without omitting the derivative step,

$$\begin{aligned} 0 &= (\nabla^2 + k^2)\left[\psi(x,y)e^{-ik_z z}\right] \\ &= \left[\nabla_\perp^2\psi + (-ik_z)^2\psi + k^2\psi\right] e^{-ik_z z} \\ &= \left[\nabla_\perp^2 + (k^2 - k_z^2)\right]\psi\, e^{-ik_z z}. \end{aligned}$$

Since the exponential is nonzero and $k_c^2 = k^2 - k_z^2$, the longitudinal field components must therefore satisfy the two-dimensional Helmholtz equation

$$\left(\nabla_\perp^2 + k_c^2\right)\psi = 0, \tag{2.3.6}$$

where ψ represents either E_z^0 or H_z^0. The longitudinal fields can be determined from this equation based on the specific matching conditions, after which the transverse fields can be obtained using (2.3.2).

Solving Waveguide Problems (cont.)

For a waveguide constructed from a PEC, referring to the geometry shown in Fig. 2.5, the tangential component of the electric field must be zero at the waveguide walls:

$$\left(\hat{n} \times \vec{E}\right)\Big|_W = 0, \tag{2.3.7}$$

where the subscript W denotes the surface of the waveguide and $\hat{n}$ is the normal unit vector pointing outwards. For a TM mode, the longitudinal direction $\hat{z}$ is tangent to the wall, so the PEC condition immediately requires

$$E_z^0\Big|_W = 0, \tag{2.3.8}$$

which is a Dirichlet boundary condition. Because E_z^0 is identically zero along the boundary, its tangential derivative also vanishes there; the TM transverse field $\vec{E}_\perp^0 \propto \vec{\nabla}_\perp E_z^0$ is therefore normal to the wall, completing the PEC condition. For a TE mode, let $\hat{\tau} = \hat{z} \times \hat{n}$ be the cross-sectional tangent. Then

$$E_\tau^0 = \vec{E}_\perp^0 \cdot \hat{\tau} = \frac{i\omega\mu}{k_c^2}(\hat{z} \times \vec{\nabla}_\perp H_z^0) \cdot (\hat{z} \times \hat{n}) = \frac{i\omega\mu}{k_c^2}\,\partial_n H_z^0.$$

Consequently the PEC condition requires

$$\left(\partial_n H_z^0\right)\Big|_W = 0, \tag{2.3.9}$$

which is a Neumann boundary condition.

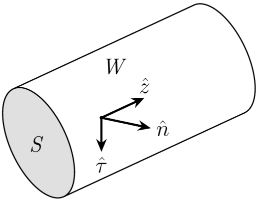

Figure 2.5: Waveguide cross-section notation. The corrected local orientation is $\hat{\tau} = \hat{z} \times \hat{n}$, with $\hat{n}$ outward from the wall and $\hat{z}$ along the guide.

Solving Waveguide Problems

Imposing these boundary conditions makes k_c^2 a discrete eigenvalue of $-\nabla_\perp^2$; k_c is its nonnegative square root. The Neumann problem has a constant $k_c = 0$ scalar eigenfunction, but it is not a physical TE_{00} mode. Indeed, the longitudinal component of Faraday's law is $(\vec{\nabla}_\perp \times \vec{E}_\perp^0) \cdot \hat{z} = -i\omega\mu H_z^0$. Integrating it over the cross-section and applying Stokes' theorem gives

$$-i\omega\mu \int_S H_z^0 \, ds = \oint_C \vec{E}_\perp^0 \cdot \hat{\tau} \, dl = 0$$

by the PEC boundary condition. A constant H_z^0 must therefore be zero. For the following cutoff relations, consider a nontrivial TE or TM eigenmode with $k_c > 0$, and assume a lossless nondispersive filling so that μ and ϵ are positive constants. Its k_c is related to its cutoff frequency f_c by

$$k_c = \omega_c \sqrt{\mu\epsilon}. \tag{2.3.10}$$

Equivalently,

$$f_c = \frac{\omega_c}{2\pi} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}}. \tag{2.3.11}$$

Solving Waveguide Problems (cont.)

From (2.3.3), with the branch chosen to propagate or decay in the $+z$ direction,

$$k_z = \begin{cases} k_c\sqrt{(f/f_c)^2 - 1} = k\sqrt{1-(f_c/f)^2}, & f > f_c, \\ 0, & f = f_c, \\ -ik_c\sqrt{1-(f/f_c)^2} = -ik\sqrt{(f_c/f)^2-1}, & 0 < f < f_c. \end{cases} \tag{2.3.12}$$

A propagating solution for that TE or TM mode exists only if k_z is real and nonzero, which requires $f > f_c$; each such mode therefore has a high-pass cutoff. A TEM mode, when the conductor topology permits one, instead has $k_c = f_c = 0$ and $k_z = k$: it has no nonzero cutoff, and the ratio forms containing f/f_c above are not used for it.
For the TE or TM mode with $k_c > 0$, write $k_z = \beta - i\alpha$ on the branch that decays in the $+z$ direction. In a lossless guide above cutoff, $\alpha = 0$ and $k_z = \beta > 0$. In a lossless guide below cutoff, $\beta = 0$ and $k_z = -i\alpha$ with $\alpha = \sqrt{k_c^2 - k^2}$; losslessness of the material does not make a below-cutoff field propagating. With $v = 1/\sqrt{\mu\epsilon}$ and the modal $\omega_c = vk_c$ constant under the stated nondispersive assumption, the guided wavelength above cutoff is

$$\lambda_z = \frac{2\pi}{\beta} = \frac{\lambda}{\sqrt{1-(f_c/f)^2}} > \lambda, \tag{2.3.13}$$

Solving Waveguide Problems (cont.)

where $\lambda = 2\pi/k$ is the wavelength of a plane wave in an unbounded medium of the same material. The phase velocity is then calculated by

$$v_{pz} = \frac{\omega}{\beta} = \frac{v}{\sqrt{1-(f_c/f)^2}} > v \tag{2.3.14}$$

and the group velocity by explicit differentiation of the dispersion relation. Writing $\beta = (\omega/v)\sqrt{1-(\omega_c/\omega)^2}$ and recalling that $\omega > 0$ above cutoff, the radical first simplifies as

$$\beta = \frac{\omega}{v}\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2} = \frac{\omega}{v}\,\frac{\sqrt{\omega^2-\omega_c^2}}{\omega} = \frac{1}{v}\sqrt{\omega^2-\omega_c^2}.$$

Its ω-derivative then follows by the chain rule,

$$\frac{d\beta}{d\omega} = \frac{1}{v}\,\frac{d}{d\omega}\left(\omega^2-\omega_c^2\right)^{1/2} = \frac{1}{v}\,\frac{\omega}{\sqrt{\omega^2-\omega_c^2}}.$$

Solving Waveguide Problems (cont.)

Inverting this slope gives the group velocity

$$\begin{aligned} v_{gz} &= \frac{d\omega}{d\beta} = \frac{1}{d\beta/d\omega} = \frac{1}{\frac{d}{d\omega}\left[\frac{\omega}{v}\sqrt{1-(\omega_c/\omega)^2}\right]} \\ &= \frac{v}{d(\sqrt{\omega^2-\omega_c^2})/d\omega} = \frac{v}{\omega/\sqrt{\omega^2-\omega_c^2}} \\ &= v\sqrt{1-(f_c/f)^2} < v, \end{aligned} \tag{2.3.15}$$

where $v = \omega/k = 1/\sqrt{\mu\epsilon}$ is the phase velocity of a plane wave in an unbounded sample of the same nondispersive material. Under these stated assumptions, the waveguide phase velocity is higher than v while its group velocity is lower than v; neither inequality implies superluminal information transfer. Their product is

$$v_{pz} v_{gz} = v^2. \tag{2.3.16}$$

Characteristics of Waveguide Modes

This section introduces the concept of modal orthogonality in waveguides. We will establish that the various modes within a waveguide form an orthogonal set of solutions. This orthogonality, which is a direct consequence of Green's identities, is demonstrated with respect to a defined inner product.

A. Green's Identities. Green's identities are fundamental results in vector calculus derived from the divergence theorem. For two scalar fields, f and g, the divergence theorem (1.1.9) leads to:

$$\int_V \vec{\nabla} \cdot \left(g \vec{\nabla} f \right) dv = \oint_S g \vec{\nabla} f \cdot d\vec{s}. \tag{2.3.17}$$

Expanding the volume integrand on the left-hand side, we obtain Green's first identity:

$$\int_V \left(g \nabla^2 f + \vec{\nabla} g \cdot \vec{\nabla} f \right) dv = \oint_S g \vec{\nabla} f \cdot d\vec{s} = \oint_S g \partial_n f ds, \tag{2.3.18}$$

where n denotes the outward normal direction. In two dimensions, this becomes:

$$\int_S \left(g \nabla_\perp^2 f + \vec{\nabla}_\perp g \cdot \vec{\nabla}_\perp f \right) ds = \oint_C g \vec{\nabla}_\perp f \cdot \hat{n} \, dl = \oint_C g \partial_n f \, dl. \tag{2.3.19}$$

Characteristics of Waveguide Modes (cont.)

Green's second identity is obtained by interchanging the roles of f and g in the first identity and subtracting the two resulting equations:

$$\begin{aligned}\int_V \left(g\nabla^2 f - f\nabla^2 g\right) dv &= \oint_S \left(g\vec{\nabla} f - f\vec{\nabla} g\right) \cdot d\vec{s} \\ &= \oint_S \left(g\partial_n f - f\partial_n g\right) ds.\end{aligned} \tag{2.3.20}$$

In two dimensions, this takes the form:

$$\begin{aligned}\int_S \left(g\nabla_\perp^2 f - f\nabla_\perp^2 g\right) ds &= \oint_C \left(g\vec{\nabla}_\perp f - f\vec{\nabla}_\perp g\right) \cdot \hat{n}\, dl \\ &= \oint_C \left(g\partial_n f - f\partial_n g\right) dl.\end{aligned} \tag{2.3.21}$$

Characteristics of Waveguide Modes (cont.)

Another useful relation can be derived from Green's first identity. By interchanging f and g in (2.3.18) and adding the resulting equations, we obtain:

$$\int_V \vec{\nabla} g \cdot \vec{\nabla} f \, dv = \frac{1}{2}\left[\oint_S (g\partial_n f + f\partial_n g)\, ds - \int_V \left(f\nabla^2 g + g\nabla^2 f\right) dv\right]. \tag{2.3.22}$$

In two dimensions, we have

$$\int_S \vec{\nabla}_\perp g \cdot \vec{\nabla}_\perp f \, ds = \frac{1}{2}\left[\oint_C (g\partial_n f + f\partial_n g)\, dl - \int_S \left(f\nabla_\perp^2 g + g\nabla_\perp^2 f\right) ds\right]. \tag{2.3.23}$$

B. Modal Orthogonality. For the lossless PEC guide considered here, and for the common chosen $+z$ propagation branch, the transverse operator $-\nabla_\perp^2$ with either Dirichlet or Neumann boundary data is self-adjoint. Its spectrum on a bounded

Characteristics of Waveguide Modes (cont.)

cross-section is countable, so one index is sufficient for an arbitrary cross-section; paired indices used for rectangles or circles are geometry-specific. Let ψ_m denote a scalar eigenfunction for either the TM or TE family:

$$\left(\nabla_\perp^2 + k_{c,m}^2\right)\psi_m = 0, \qquad m = 1, 2, \ldots, \tag{2.3.24}$$

with, respectively,

$$\psi_m|_W = 0 \qquad \text{(TM/Dirichlet)}, \tag{2.3.25a}$$

$$\partial_n \psi_m|_W = 0 \qquad \text{(TE/Neumann)}. \tag{2.3.25b}$$

The Neumann zero eigenfunction is excluded for the Maxwell reason given above. Because phasor fields can be complex, the relevant inner product is Hermitian,

$$\langle f, g\rangle_S = \int_S f\, g^*\, ds,$$

not the unconjugated product. The eigenvalues $k_{c,m}^2$ are real. Multiply the equation for ψ_m by ψ_n^*, multiply the conjugate equation for ψ_n by ψ_m, and subtract:

$$\psi_n^* \nabla_\perp^2 \psi_m - \psi_m \nabla_\perp^2 \psi_n^* = \left(k_{c,n}^2 - k_{c,m}^2\right)\psi_m \psi_n^*. \tag{2.3.26}$$

Characteristics of Waveguide Modes (cont.)

Green's second identity then gives

$$\oint_C \left(\psi_n^* \partial_n \psi_m - \psi_m \partial_n \psi_n^*\right) dl = \left(k_{c,n}^2 - k_{c,m}^2\right) \int_S \psi_m \psi_n^* ds. \tag{2.3.27}$$

Within one scalar family, both terms on the boundary vanish: $\psi_m = \psi_n = 0$ for Dirichlet data, or $\partial_n \psi_m = \partial_n \psi_n = 0$ for Neumann data. Hence modes with different eigenvalues are orthogonal. If an eigenvalue is degenerate, orthogonality does not follow from division by $k_{c,n}^2 - k_{c,m}^2$; instead, an orthogonal basis is chosen within that finite-dimensional eigenspace. After doing so, write

$$\int_S \psi_m \psi_n^* ds = N_m \delta_{mn}, \qquad N_m > 0. \tag{2.3.28}$$

In particular, distinct longitudinal fields in each family obey

$$\int_S \vec{E}_{z,m}^0 \cdot \vec{E}_{z,n}^{0*} \, ds = 0 \qquad \text{for distinct TM modes,} \tag{2.3.29a}$$

$$\int_S \vec{H}_{z,m}^0 \cdot \vec{H}_{z,n}^{0*} \, ds = 0 \qquad \text{for distinct TE modes.} \tag{2.3.29b}$$

Characteristics of Waveguide Modes (cont.)

Green's first identity gives the corresponding gradient normalization without omitting the boundary step:

$$\begin{aligned}\int_S \vec{\nabla}_\perp \psi_m \cdot \vec{\nabla}_\perp \psi_n^* \, ds &= \oint_C \psi_n^* \partial_n \psi_m \, dl - \int_S \psi_n^* \nabla_\perp^2 \psi_m \, ds \\ &= 0 + k_{c,m}^2 \int_S \psi_m \psi_n^* \, ds \\ &= k_{c,m}^2 N_m \delta_{mn}.\end{aligned}$$

Thus

$$\int_S \vec{\nabla}_\perp \psi_m \cdot \vec{\nabla}_\perp \psi_n^* \, ds = 0 \qquad (m \neq n). \tag{2.3.30}$$

The identity

$$(\hat{z} \times \vec{a}) \cdot (\hat{z} \times \vec{b}^*) = \vec{a} \cdot \vec{b}^* \qquad (\vec{a}, \vec{b} \perp \hat{z})$$

shows from (2.3.4) and (2.3.5) that this proves transverse-field orthogonality for distinct modes within the TM family and within the TE family.
It remains to check a TM–TE pair rather than assume that the two different boundary-value families are scalar-orthogonal. Let $u_m = E_{z,m}^0$ be a Dirichlet TM

Characteristics of Waveguide Modes (cont.)

eigenfunction and $v_n = H^0_{z,n}$ a Neumann TE eigenfunction. Since mixed partial derivatives commute,

$$\vec{\nabla}_\perp \cdot (\hat{z} \times \vec{\nabla}_\perp v_n^*) = 0.$$

The product rule and the two-dimensional divergence theorem therefore give

$$\begin{aligned}\int_S \vec{\nabla}_\perp u_m \cdot (\hat{z} \times \vec{\nabla}_\perp v_n^*)\, ds &= \int_S \vec{\nabla}_\perp \cdot \left[u_m(\hat{z} \times \vec{\nabla}_\perp v_n^*)\right] ds \\ &= \oint_C u_m(\hat{z} \times \vec{\nabla}_\perp v_n^*) \cdot \hat{n}\, dl = 0,\end{aligned} \tag{2.3.31}$$

because $u_m = 0$ on the PEC boundary. Interchanging the two gradients or rotating both of them by $\hat{z}\times$ changes at most a sign, so all TM–TE transverse overlaps required below also vanish. Combining the same-family and cross-family results,

$$\int_S \vec{E}^0_{\perp,m} \cdot \vec{E}^{0*}_{\perp,n}\, ds = 0, \tag{2.3.32a}$$

$$\int_S \vec{H}^0_{\perp,m} \cdot \vec{H}^{0*}_{\perp,n}\, ds = 0, \tag{2.3.32b}$$

for two distinct modes in the orthogonal modal basis.
The physically relevant power overlap also requires conjugation. Define

$$P_{mn} = \frac{1}{2} \int_S \left(\vec{E}^0_{\perp,m} \times \vec{H}^{0*}_{\perp,n} \right) \cdot \hat{z} \, ds.$$

For transverse vectors, the identities

$$\begin{aligned}
[\vec{a} \times (\hat{z} \times \vec{b}^*)] \cdot \hat{z} &= \vec{a} \cdot \vec{b}^*, \\
[(\hat{z} \times \vec{a}) \times \vec{b}^*] \cdot \hat{z} &= -\vec{a} \cdot \vec{b}^*, \\
[(\hat{z} \times \vec{a}) \times (\hat{z} \times \vec{b}^*)] \cdot \hat{z} &= (\vec{a} \times \vec{b}^*) \cdot \hat{z}
\end{aligned}$$

reduce every TM–TM and TE–TE cross-power term to (2.3.30), and every TM–TE term to (2.3.31). Consequently

$$P_{mn} = P_m \delta_{mn}. \tag{2.3.33}$$

The diagonal term P_m is generally nonzero for a propagating mode; the earlier unconjugated statement that the integral is zero "in general" would incorrectly set a mode's own carried power to zero. In an ideal uniform lossless guide this orthogonal basis makes different modal amplitudes evolve independently. Degenerate eigenspaces must first be orthogonalized, while wall loss or longitudinal/material perturbations lead to non-Hermitian or coupled-mode formulations rather than the Hermitian proof above.

Further Reading

Advanced topics such as wave propagation in inhomogeneous, layered, or anisotropic media are presented in Ishimaru's text [12] and Yeh's monograph *Optical Waves in Layered Media* [30]. For a more detailed discussion of inhomogeneous media, Chew's book [5] provides an excellent treatment and also serves as a solid general introduction to electromagnetic theory.

The angular spectrum representation (plane-wave expansion) is discussed in independent chapters by Novotny and Hecht [21], Pathak [23], and in depth in Clemmow's monograph *The Plane Wave Spectrum Representation of Electromagnetic Fields* [6]. Applications of this formulation to planar near-field measurement techniques are presented in Gregson et al. [10].

For guided-wave propagation, both Ishimaru [12] and Zangwill [31] offer a concise overview in relevant chapters, whereas Collin's authoritative text *Field Theory of Guided Waves* [7] provides a comprehensive and rigorous treatment of waveguide transmission.

Problems

1. Verify (2.1.9) with gradient, divergence and curl in Cartesian coordinate system.
2. Derive (2.2.7) and (2.2.8).
3. Rewrite the total-internal-reflection field (2.2.32) in the form (2.2.5), using the real medium wavenumber $k = k_2$. Identify $\vec{k}' = (k_1/k_2)\sin\theta_i\,\hat{y}$ and $\vec{k}'' = \sinh\delta_t\,\hat{x}$, and then verify explicitly that they satisfy (2.2.3a) and (2.2.3b).
4. Derive (2.3.2) and express them in Cartesian coordinates.
5. For a mode with $k_z \neq 0$, show that for a TM mode we have

$$\vec{E}_\perp^0 = -\frac{k_z}{\omega\epsilon}\hat{z}\times\vec{H}_\perp^0,$$
$$\vec{H}_\perp^0 = \frac{\omega\epsilon}{k_z}\hat{z}\times\vec{E}_\perp^0,$$

and for TE mode, we have

$$\vec{E}_\perp^0 = -\frac{\omega\mu}{k_z}\hat{z}\times\vec{H}_\perp^0,$$
$$\vec{H}_\perp^0 = \frac{k_z}{\omega\mu}\hat{z}\times\vec{E}_\perp^0.$$

Problems (cont.)

6. Show that a nontrivial TEM mode cannot exist in a homogeneous hollow waveguide of arbitrary simply connected cross-section bounded by a single PEC conductor. Explain why this proof does not apply to a multiply connected guide such as a coaxial line.

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